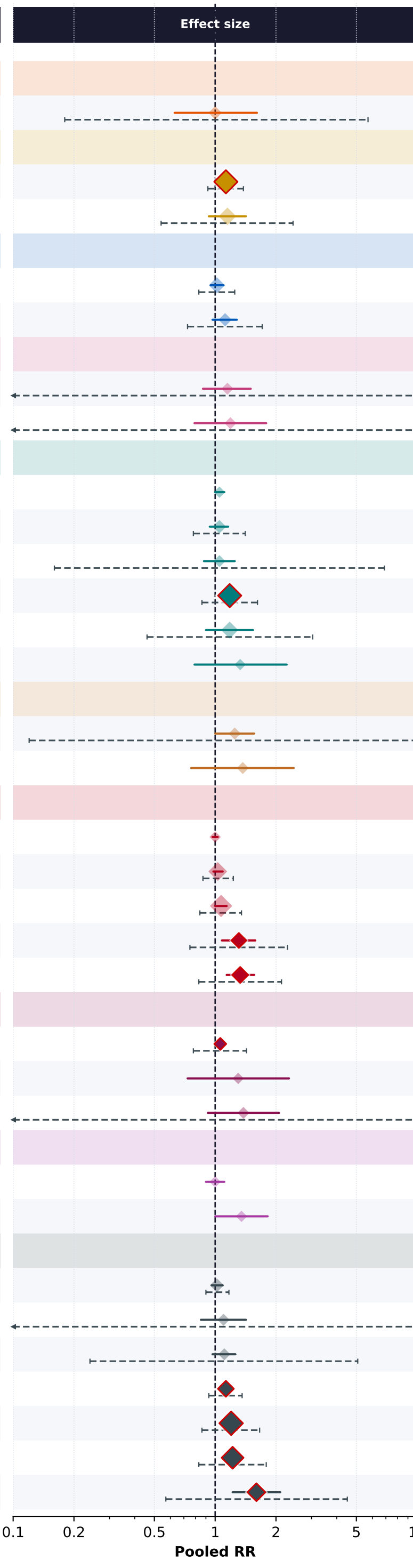
# Exposures positively associated with prostate cancer risk

| Exposure | Studies | N | Cases | Pooled RR (95% CI) |
|---|---|---|---|---|
| ***CAROTENOIDS*** | | | | |
| **Lutein** | 5 | 3,128 | 1,292 | *1.00 (0.63-1.61)* |
| ***VITAMINS A, C, D, E, K*** | | | | |
| **Vitamin D** | 28 | 551,351 | - | *1.13 (1.07-1.19)* |
| **Vitamin A** | 13 | 70,012 | 17,837 | *1.15 (0.93-1.42)* |
| ***B VITAMINS*** | | | | |
| **Folic acid** | 10 | 157,607 | - | *1.02 (0.95-1.10)* |
| **Vitamin B12** | 5 | 5,961 | 2,681 | *1.12 (0.97-1.28)* |
| ***ANTIOXIDANTS*** | | | | |
| **Antioxidants** | 3 | 56,601 | 7,271 | *1.15 (0.87-1.50)* |
| **Coenzyme Q10** | 3 | 36,416 | 2,065 | *1.19 (0.79-1.79)* |
| ***MINERALS & TRACE ELEMENTS*** | | | | |
| **Copper** | 2 | 43,360 | 6,174 | *1.05 (1.00-1.11)* |
| **Iron** | 5 | 184,031 | 11,578 | *1.05 (0.94-1.16)* |
| **Magnesium** | 3 | 44,056 | 6,498 | *1.05 (0.88-1.25)* |
| **Calcium** | 28 | 1,049,042 | - | *1.18 (1.10-1.28)* |
| **Zinc** | 12 | 96,718 | 7,983 | *1.18 (0.90-1.54)* |
| **Mineral supplements** | 2 | 57,251 | 1,958 | *1.33 (0.79-2.26)* |
| ***FERMENTED FOODS & PROBIOTICS*** | | | | |
| **Yogurt** | 3 | 201,324 | 7,680 | *1.25 (1.00-1.56)* |
| **Fermented foods** | 3 | 3,460 | 339 | *1.37 (0.76-2.45)* |
| ***FATTY ACIDS & LIPIDS*** | | | | |
| **Docosapentaenoic acid (omega-3)** | 2 | 143,344 | 7,412 | *1.00 (0.97-1.03)* |
| **Omega-3 fatty acids** | 16 | 896,938 | 34,805 | *1.03 (0.98-1.09)* |
| **Polyunsaturated Fat (PUFA)** | 25 | 953,107 | - | *1.07 (1.00-1.14)* |
| **Alpha-linolenic acid** | 8 | 557,816 | 13,151 | *1.31 (1.08-1.58)* |
| **Monounsaturated Fat (MUFA)** | 10 | 257,768 | - | *1.33 (1.14-1.56)* |
| ***METABOLITES & AMINO ACIDS*** | | | | |
| **Glucosamine** | 3 | 265,129 | 16,136 | *1.06 (1.01-1.11)* |
| **Betaine** | 2 | 48,588 | 868 | *1.30 (0.73-2.32)* |
| **Choline** | 3 | 8,815 | 1,545 | *1.38 (0.92-2.07)* |
| ***DIET*** | | | | |
| **Mediterranean diet** | 2 | 106,146 | 7,476 | *1.00 (0.90-1.11)* |
| **Western pattern diet** | 2 | 44,462 | 1,702 | *1.35 (1.00-1.82)* |
| ***OTHER*** | | | | |
| **Multivitamin** | 6 | 397,821 | 7,673 | *1.02 (0.96-1.09)* |
| **Ultra-processed food intake** | 3 | 4,983 | 2,932 | *1.10 (0.85-1.42)* |
| **Whole-grain intake** | 3 | 79,458 | 7,209 | *1.11 (0.97-1.26)* |
| **Processed meat intake** | 10 | 1,091,655 | 65,703 | *1.13 (1.05-1.21)* |
| **Alcohol** | 29 | 1,934,957 | - | *1.20 (1.10-1.30)* |
| **Dairy** | 24 | 739,672 | 33,998 | *1.22 (1.12-1.34)* |
| **Red meat** | 14 | 974,550 | 57,630 | *1.60 (1.22-2.10)* |




Exposure group

Carotenoids · Vitamins A, C, D, E, K · B Vitamins · Antioxidants · Minerals & Trace Elements · Fermented Foods & Probiotics · Fatty Acids & Lipids · Metabolites & Amino Acids · Diet · Other

*RR > 1 = positively associated with prostate cancer risk. |◆ pooled RR (size reflects number of studies) |◇ red outline = statistically significant (95% CI excludes 1.0) |◆ faded = not statistically significant | dashed line below CI = 95% prediction interval (where available) | arrowhead = interval continues beyond the axis*

Include subexposures: Yes